Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Esther Oh, M.D., Ph.D.	Email	tipa@jhu.edu
Institute Name	Johns Hopkins University School of Medicine			
Address	5501 Hopkins Bayview Circle, Rm 1B.76, Baltimore, MD 21224			

2. Technology Overview

Title of Proposed Technology	AI-based, Language- and Modality-Agnostic Digital Platforms for Early Cognitive Impairment Screening			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio & Medical		Information and Communications	
	Medium Category	**Small Category**	**Medium Category**	**Small Category**
	Medical Information and Systems	Remote and Home Healthcare Devices	Software	Software Solutions
Concept of Proposed Technology	Early detection of mild cognitive impairment (MCI) and dementia remains a critical unmet need, particularly in primary care and community-based settings where access to specialized neuropsychological testing is limited. Conventional screening tools are often time-consuming, resource-intensive, or insufficiently sensitive to early-stage cognitive decline. This joint R&D seeks to support the development of AI-driven, speech- or behavior-based digital screening technologies that: • Enable non-invasive, rapid assessment of cognitive function; • Operate independently of a specific language, culture, or clinical environment, and can be efficiently adapted across regions; • Are deployable via widely accessible consumer devices such as smartphones, tablets, or telehealth platforms; • Support scalable population-level screening and longitudinal monitoring. Proposed technologies may include, but are not limited to:			

(max 10 pages)

	• AI models that analyze speech, audio, or other digital biomarkers associated with cognitive function; • Language-independent or cross-linguistic architectures that minimize redevelopment for international deployment; • Digital assessment frameworks optimized for older adults and non-specialist care settings; • Platforms that facilitate early referral, monitoring, or triage rather than definitive diagnosis.
Rationale for Joint R&D Project	Despite growing interest in digital cognitive screening, successful clinical translation requires rigorous validation, culturally and linguistically appropriate task design, and real-world clinical evidence, particularly within the U.S. healthcare system. The rationale for a Korea–US joint R&D framework includes: • Korean startups/SMEs bring strengths in AI algorithm development, mobile platform engineering, and rapid product iteration, but may face limitations in conducting large-scale U.S.-based clinical studies. • U.S. academic medical centers offer access to diverse patient populations, standardized clinical assessment frameworks, and expertise in clinical trial execution and regulatory-aligned evidence generation. Through joint collaboration, teams can: • Ensure that digital screening technologies are clinically meaningful and culturally appropriate for English-speaking and diverse populations; • Generate high-quality datasets necessary for algorithm validation and generalization; • De-risk global deployment through early alignment with U.S. clinical and regulatory expectations.
Key Words	1. Speech-Based Cognitive Screening 2. Language-Independent AI Algorithm 3. English Speech Data Collection for Algorithm Training 4. Early Detection of Cognitive Impairment 5. AI-Based Dementia Screening Solution

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)
R&D Objectives	Proposals submitted under this joint R&D program should aim to achieve the following high-level objectives: 1. Development of AI-based digital screening technologies capable of detecting early cognitive impairment; 2. Adaptation or validation of these technologies for English-speaking and U.S. clinical contexts; 3. Demonstration of clinical performance and usability through structured studies; 4. Preparation for future regulatory clearance and commercialization as digital medical technologies. Projects are expected to advance from clinical feasibility and pilot validation (TRL 3–4) toward deployable and clinically validated solutions (TRL 5–7).			
R&D Target Specifications	1. Language-Adaptable Digital Cognitive Tasks • Develop clinically appropriate speech- or behavior-based digital tasks for English-speaking older adults. • Establish a localization framework extensible to additional languages and regions. 2. Clinically Annotated English-Language Dataset • Generate a clinically labeled English-language dataset across the cognitive spectrum through collaboration with U.S. academic medical centers. • Ensure dataset sufficiency for algorithm training, validation, and generalizability assessment. 3. Algorithm Performance and Clinical Validity • Demonstrate clinically meaningful performance in distinguishing cognitive impairment using English-language data. • Maintain performance comparable to or exceeding prior validated implementations in other populations. 4. System Robustness and Reliability • Ensure high system stability and data integrity during clinical studies and real-world operation. • Demonstrate readiness for routine clinical or community deployment. 5. Deployment and Commercialization Readiness • Deliver a commercialization-ready digital cognitive screening solution for English-speaking settings. • Demonstrate usability and feasibility through pilot or proof-of-concept studies with U.S. partners.			
R&D Plan	While detailed plans should be proposed by applicants, the anticipated scope of R&D may include: • Localization or adaptation of digital cognitive tasks for target populations; • Clinical data collection and labeling in collaboration with U.S. research partners; • Training and validation of AI models using clinically annotated datasets; • Assessment of system stability, usability, and acceptability in real-world settings.			

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Clinical study design, ethical approval, and execution; • Recruitment and classification of study participants; • Clinical interpretation, validation, and scientific dissemination.
Expected R&R of Korean SME Partner	• Core AI algorithm and platform development; • System engineering, data processing, and internal validation; • Productization, intellectual property generation, and market planning.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technological Outcomes • Clinically validated digital screening platforms for early cognitive impairment; • Reusable and scalable AI architectures applicable across languages and regions; • Strengthened Korea–US collaboration in digital cognitive health. Clinical and Societal Impact • Improved access to early cognitive screening outside specialized hospitals; • Support for early intervention and improved care pathways; • Reduced burden on resource-intensive diagnostic workflows. Commercial Impact • Creation of globally deployable digital health products; • Acceleration of international market entry for Korean AI healthcare technologies; • Long-term opportunities for licensing, partnerships, and ecosystem growth.

6. References

1) Choi, Y., et al. “Deep-learning analysis of speech using mel-spectrograms for the assessment of mild cognitive impairment and Alzheimer’s disease.” *Journal of Alzheimer’s Disease*

2) Yoon, B., et al. “Clinical Validation of a Voice-Based AI Tool for Screening Cognitive Impairment: A Prospective Multicenter Study.” *Neurological Sciences*